

Second quantization: creation and annihilation operators, and Wick's theorem

FYS4480/FYS9480 – Week 35, August 27–28, 2026

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Thursday August 27: reminder of week 34, and creation and annihilation operators

Friday August 28: one- and two-body operators in second quantization, and Wick's theorem

Reading: these slides, chapter 3 of the lecture notes (sections 3.1–3.2 and 3.5–3.11),

Szabo and Ostlund chapter 2. Code: `WeeklySlides/week35.ipynb` and

`BookPrograms/chapter03/wick.py`

Plan for week 35

Thursday August 27, 2×45 min

- Reminder of last Friday's lecture: the variational principle, expectation values with Slater determinants, direct and exchange terms, and the energy functional
- Why first quantization does not scale: the permutation bookkeeping
- Second quantization: creation and annihilation operators and their anticommutation relations
- The number operator and one-body operators in second quantized form

Friday August 28, 2×45 min

- Two-body operators in second quantization: the results (derivations in the book, section 3.2)
- Normal ordering and contractions
- Two, three and four operators: building up Wick's theorem
- Wick's theorem and its proof by induction
- Exercise session: exercise set for week 35

Thursday, lecture 1: Reminder – expectation values and the energy functional

Where week 34 ended

Last Friday we computed quantum mechanical expectation values with a Slater determinant, by hand, in first quantization. The results we carry into this week:

- The Hamiltonian is a one-body part plus a two-body interaction, $\hat{H} = \hat{H}_0 + \hat{H}_I$.
- The variational principle: any normalised trial state gives an energy above the exact ground state.
- In a Slater determinant, $\langle \hat{H}_0 \rangle$ is a sum over occupied orbitals; $\langle \hat{H}_I \rangle$ has a **direct** and an **exchange** term.
- The energy is a functional of the orbitals; minimising it will give Hartree-Fock.

Today's first 45 minutes re-derive nothing: they collect these results, and pay attention to *how much work* the derivations cost. That cost is the motivation for the rest of the week.

The Hamiltonian and the single-particle basis (week 34)

Throughout the course the interacting part is approximated by a two-body interaction,

$$\hat{H} = \hat{H}_0 + \hat{H}_I = \sum_{i=1}^N \hat{h}_0(x_i) + \sum_{i < j}^N \hat{v}(r_{ij}), \quad \hat{h}_0(x_i) = \hat{t}(x_i) + \hat{u}_{\text{ext}}(x_i),$$

with x_i the coordinates and spin of particle i . The single-particle functions are eigenfunctions of the one-body part,

$$\hat{h}_0(x_i)\psi_\alpha(x_i) = \varepsilon_\alpha\psi_\alpha(x_i),$$

and we keep the particle-hole labelling convention adopted last week:

$i, j, k, l, \dots \leq F$	occupied (hole) states below the Fermi level F ,
$a, b, c, d, \dots > F$	unoccupied (particle) states,
$p, q, r, s, \dots$	general single-particle states.

Slater determinants and the antisymmetrisation operator (week 34)

The N -fermion ansatz is the Slater determinant, which we wrote as the antisymmetrised Hartree function,

$$\Phi = \frac{1}{\sqrt{N!}} \sum_P (-)^P \hat{P} \Phi_H = \sqrt{N!} \hat{A} \Phi_H, \quad \Phi_H(x_1, \dots, x_N) = \psi_\alpha(x_1) \psi_\beta(x_2) \cdots \psi_\nu(x_N),$$

with the antisymmetrisation operator

$$\hat{A} = \frac{1}{N!} \sum_P (-)^P \hat{P}, \quad [\hat{H}_0, \hat{A}] = [\hat{H}_I, \hat{A}] = 0, \quad \hat{A}^2 = \hat{A}.$$

- Interchanging two particles interchanges two columns: the sign flips (antisymmetry).
- Two particles in the same orbital give two identical rows: the determinant vanishes (Pauli).
- A unitary rotation among the occupied orbitals changes the determinant by a phase only.

The variational principle (week 34)

For any normalised trial function Φ ,

$$E_0 \leq E[\Phi] = \int \Phi^* \hat{H} \Phi \, d\tau,$$

proved in one line by expanding Φ in the exact eigenstates: $E[\Phi] = \sum_k |c_k|^2 E_k \geq E_0$.

This one inequality is the engine of most of the course

Trial function = one Slater determinant, minimise $\Rightarrow$ Hartree-Fock. Linear combination of determinants $\Rightarrow$ configuration interaction. Parametrised quantum circuit $\Rightarrow$ the variational quantum eigensolver. Correlated trial function sampled stochastically $\Rightarrow$ variational Monte Carlo.

What we need, over and over again, is therefore the **expectation value of $\hat{H}$ in determinant-based states** – and that is precisely the computation whose bookkeeping we now recall.

The one-body part: only the identity permutation survives (week 34)

Using $[\hat{H}_0, \hat{A}] = 0$ and $\hat{A}^2 = \hat{A}$,

$$\int \Phi^* \hat{H}_0 \Phi \, d\tau = \sum_{i=1}^N \sum_p (-)^p \int \Phi_H^* \hat{h}_0(x_i) \hat{P} \Phi_H \, d\tau.$$

$\hat{h}_0(x_i)$ touches one coordinate. Any permutation that moves a particle leaves at least one overlap $\langle \mu | \nu \rangle$ between two different orbitals, which vanishes. Only the identity survives:

$$\int \Phi^* \hat{H}_0 \Phi \, d\tau = \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle.$$

One page of argument in the lecture notes – for the *simplest* operator there is.

The two-body part: direct and exchange (week 34)

For $\hat{v}(r_{ij})$, which touches two coordinates, **two** permutations survive – the identity and the transposition $\hat{P}_{ij}$:

$$\int \Phi^* \hat{H}_I \Phi \, d\tau = \sum_{i < j}^N \int \Phi_H^* \hat{v}(r_{ij}) (1 - \hat{P}_{ij}) \Phi_H \, d\tau = \frac{1}{2} \sum_{\mu\nu}^N \left[\langle \mu\nu | \hat{v} | \mu\nu \rangle - \langle \mu\nu | \hat{v} | \nu\mu \rangle \right].$$

- The first term is the **direct** (Hartree) term: classical electrostatics of two charge clouds.
- The second is the **exchange** (Fock) term: no classical counterpart, purely a consequence of antisymmetry – the trace left in the energy by $\hat{P}_{ij}$.

With the **antisymmetrised matrix element**

$$\langle \mu\nu | \hat{v} | \mu\nu \rangle_{AS} = \langle \mu\nu | \hat{v} | \mu\nu \rangle - \langle \mu\nu | \hat{v} | \nu\mu \rangle,$$

we have $\langle \mu\mu | \hat{v} | \mu\mu \rangle_{AS} = 0$: no self-interaction.

The energy functional (week 34)

Collecting both parts, with i, j running over the occupied orbitals,

$$E[\Phi] = \sum_{i=1}^N \langle i | \hat{h}_0 | i \rangle + \frac{1}{2} \sum_{i,j=1}^N \langle ij | \hat{v} | ij \rangle_{\text{AS}},$$

and after expanding the orbitals in a fixed basis, $\psi_i = \sum_{\alpha} C_{i\alpha} \phi_{\alpha}$,

$$E[\Psi] = \sum_{i=1}^N \sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \frac{1}{2} \sum_{i,j=1}^N \sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{\text{AS}}.$$

Minimising this functional over the coefficients is the Hartree-Fock method – a self-consistent eigenvalue problem, previewed last week and derived properly later in the course.

Pause to think: the exchange term

Pause to think #1

(about 3 min – discuss with your neighbour)

1. In the two-body expectation value, where exactly does the *minus* sign of the exchange term come from?
2. Why is $\langle \mu\mu | \hat{v} | \mu\mu \rangle_{AS} = 0$ physically important? What goes wrong in a theory (Hartree's original one) that keeps only the direct term?

Answer to pause #1

1. From the parity of the surviving permutation: the transposition $\hat{P}_{ij}$ is odd, so it enters $\sum_p (-)^p$ with $(-1)^1$. The direct term comes with the identity (even, +). One sign per permutation parity – the same rule that defines a determinant.
2. It removes the self-interaction: the $\mu = \nu$ terms of the direct sum, where a particle feels the field generated by itself, are cancelled exactly by the exchange sum. In Hartree theory this cancellation is absent, and each electron spuriously repels itself.

The cost of doing this by hand

Every result above was obtained by writing out the antisymmetriser, sorting through $N!$ permutations, and arguing term by term about which overlaps vanish.

- For the one-body operator: one page.
- For the two-body operator: two pages, and a factor-of-two subtlety in the pair counting.
- For matrix elements *between different* determinants – which perturbation theory, configuration interaction and coupled cluster theory need everywhere – the same argument must be repeated for every pair of determinants, with signs that depend on how the determinants differ.
- For the eight-operator strings of perturbation theory: hopeless.

This week's programme

Build the antisymmetry **once and for all** into the algebra of operators, so that permutation bookkeeping becomes operator algebra – and then automate the algebra itself (Wick's theorem).

Thursday, lecture 2: Creation and annihilation operators

Second quantization: the idea

In first quantization the state carries the antisymmetry: $\Phi(x_1, \dots, x_N)$ is a determinant, and operators are innocent. In **second quantization** (the occupation-number representation) the roles are reversed:

- states are specified by *which single-particle states are occupied* – nothing more;
- all antisymmetry lives in the *algebra* of two operators, $a_\alpha^\dagger$ and a_α , which create and annihilate a particle in the single-particle state φ_α ;
- the Slater determinant becomes a string of creation operators acting on the vacuum, and the Condon-Slater rules will become consequences of Wick's theorem.

Calculations that were barely feasible by hand for $N = 3$ become mechanical for arbitrary N . The price: we must first establish the operator algebra, carefully, once.

The creation operator

We define the fermion creation operator $a_{\alpha}^{\dagger}$ by its action on the particle vacuum $|0\rangle$ (the state with no particles) and on an antisymmetric n -particle state,

$$\begin{aligned}a_{\alpha}^{\dagger}|0\rangle &\equiv |\alpha\rangle, \\a_{\alpha}^{\dagger}|\alpha_1 \dots \alpha_n\rangle_{AS} &\equiv |\alpha\alpha_1 \dots \alpha_n\rangle_{AS},\end{aligned}$$

the second line under the condition $\alpha \neq \alpha_1, \dots, \alpha_n$: it creates an antisymmetric $(n+1)$ -particle state with φ_{α} occupied.

Every antisymmetric state is a string of creation operators on the vacuum

$$|\alpha_1 \dots \alpha_n\rangle_{AS} = a_{\alpha_1}^{\dagger} a_{\alpha_2}^{\dagger} \dots a_{\alpha_n}^{\dagger} |0\rangle.$$

This is the Slater determinant of week 34 – written without a single coordinate, permutation or determinant in sight.

Anticommutation rules for creation operators

The antisymmetry of the states does all the work. From

$$|\alpha_1 \dots \alpha_i \dots \alpha_k \dots \alpha_n\rangle_{AS} = -|\alpha_1 \dots \alpha_k \dots \alpha_i \dots \alpha_n\rangle_{AS}$$

we obtain

$$a_{\alpha_i}^\dagger a_{\alpha_k}^\dagger = -a_{\alpha_k}^\dagger a_{\alpha_i}^\dagger,$$

and from the Pauli principle $|\alpha_1 \dots \alpha_i \dots \alpha_i \dots \alpha_n\rangle_{AS} = 0$ it follows that

$$a_{\alpha_i}^\dagger a_{\alpha_i}^\dagger = 0.$$

Combining the two,

$$\{a_\alpha^\dagger, a_\beta^\dagger\} \equiv a_\alpha^\dagger a_\beta^\dagger + a_\beta^\dagger a_\alpha^\dagger = 0$$

and taking the Hermitian conjugate, with $a_\alpha = (a_\alpha^\dagger)^\dagger$,

$$\{a_\alpha, a_\beta\} = 0.$$

Antisymmetry and the Pauli principle are now *encoded in the operators*.

What does a_α do? Setting up the question

The physical interpretation of a_α follows from the matrix element

$$\langle \alpha_1 \alpha_2 \dots \alpha_n | a_\alpha | \alpha'_1 \alpha'_2 \dots \alpha'_m \rangle,$$

with both states antisymmetric. Two cases for the bra:

1. $\alpha \in \{\alpha_i\}$: then $\langle \alpha_1 \dots \alpha_n | a_\alpha = 0$ by the Pauli principle (the conjugate state would occupy α twice).
2. $\alpha \notin \{\alpha_i\}$: Hermitian conjugation of the definition of $a_\alpha^\dagger$ gives

$$\langle \alpha_1 \alpha_2 \dots \alpha_n | a_\alpha = \langle \alpha \alpha_1 \alpha_2 \dots \alpha_n |,$$

which also covers case 1 (both sides vanish). Hence

$$\langle \alpha_1 \dots \alpha_n | a_\alpha | \alpha'_1 \dots \alpha'_m \rangle = \langle \alpha_1 \dots \alpha_n | \alpha \alpha'_1 \dots \alpha'_m \rangle$$

non-zero only if $m = n + 1$, with sign given by the parity of the permutation relating the two orderings.

What does a_α do? The answer

Reading the matrix element for all possible kets:

$$a_\alpha \underbrace{|\alpha'_1 \alpha'_2 \dots \alpha'_{n+1}\rangle}_{\not\equiv \alpha} = 0, \quad \text{in particular} \quad a_\alpha |0\rangle = 0,$$

and if the state contains α (brought to the front by permutations),

$$a_\alpha |\alpha \alpha_1 \alpha_2 \dots \alpha_n\rangle = |\alpha_1 \alpha_2 \dots \alpha_n\rangle.$$

Interpretation

a_α **removes** a particle from the single-particle state α ; if α is not occupied the result is zero. It is called a destruction or **annihilation** operator.

Note the bookkeeping hidden in "brought to the front": if α sits at position $k+1$, moving it to the front costs $(-1)^k$. The signs have not disappeared – they have moved into the algebra, where they take care of themselves.

Pause to think: signs from the algebra

Pause to think #2

(about 3 min – discuss with your neighbour)

Let $|\alpha_1\alpha_2\alpha_3\rangle = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger a_{\alpha_3}^\dagger |0\rangle$ with all three labels distinct.

1. Compute $a_{\alpha_2}|\alpha_1\alpha_2\alpha_3\rangle$, keeping track of the sign.
2. Compute $a_{\alpha_2}^\dagger a_{\alpha_2}|\alpha_1\alpha_2\alpha_3\rangle$. What does this operator do?

Answer to pause #2

1. Move α_2 to the front first: $|\alpha_1\alpha_2\alpha_3\rangle = -|\alpha_2\alpha_1\alpha_3\rangle$ (one transposition). Then $a_{\alpha_2}|\alpha_1\alpha_2\alpha_3\rangle = -a_{\alpha_2}|\alpha_2\alpha_1\alpha_3\rangle = -|\alpha_1\alpha_3\rangle$.
2. Acting with $a_{\alpha_2}^\dagger$ on the result: $-a_{\alpha_2}^\dagger|\alpha_1\alpha_3\rangle = -|\alpha_2\alpha_1\alpha_3\rangle = +|\alpha_1\alpha_2\alpha_3\rangle$. The state is reproduced with coefficient 1: $a_{\alpha_2}^\dagger a_{\alpha_2}$ *counts* whether α_2 is occupied – eigenvalue 1 if yes, 0 if no. This is the occupation-number operator for the state α_2 , and the two signs cancelling is why it works.

The mixed anticommutator

The remaining relation is $\{a_\alpha^\dagger, a_\beta\}$. For $\alpha = \beta$, act on any state:

$$a_\alpha^\dagger a_\alpha \underbrace{|\alpha_1 \dots \alpha_n\rangle}_{\not\equiv \alpha} = 0, \quad a_\alpha a_\alpha^\dagger \underbrace{|\alpha_1 \dots \alpha_n\rangle}_{\not\equiv \alpha} = a_\alpha |\alpha \alpha_1 \dots \alpha_n\rangle = |\alpha_1 \dots \alpha_n\rangle,$$

and with α present the roles interchange ($a_\alpha^\dagger a_\alpha$ gives the state back, $a_\alpha a_\alpha^\dagger$ kills it). In both cases the *sum* acts as the identity:

$$\{a_\alpha^\dagger, a_\alpha\} = 1.$$

For $\alpha \neq \beta$ there are three cases – the state contains both α and β , only one of them, or neither – and checking each one (see the book) gives

$$\{a_\alpha^\dagger, a_\beta\} = 0, \quad \alpha \neq \beta.$$

The algebra, complete

Second quantization for fermions: everything we will ever need

$$a_{\alpha}^{\dagger}|0\rangle = |\alpha\rangle, \quad |\alpha_1 \dots \alpha_n\rangle_{AS} = a_{\alpha_1}^{\dagger} \cdots a_{\alpha_n}^{\dagger}|0\rangle, \quad a_{\alpha}|0\rangle = 0,$$

$$\boxed{\{a_{\alpha}^{\dagger}, a_{\beta}^{\dagger}\} = \{a_{\alpha}, a_{\beta}\} = 0, \quad \{a_{\alpha}^{\dagger}, a_{\beta}\} = \delta_{\alpha\beta}}$$

- Three anticommutators replace all permutation bookkeeping of chapter 2.
- Every calculation in the rest of the course – Hartree-Fock, FCI, perturbation theory, coupled cluster – is a sequence of applications of these three lines.
- In the notebook the operators are built as matrices in a small Fock space and all three relations are verified numerically, for every pair of states.

The number operator

For $\alpha = \beta$ the operator $a_\alpha^\dagger a_\alpha$ counts the occupation of α (pause #2):

$$a_\alpha^\dagger a_\alpha |\alpha_1 \alpha_2 \dots \alpha_n\rangle = \begin{cases} 0 & \alpha \notin \{\alpha_i\}, \\ |\alpha_1 \alpha_2 \dots \alpha_n\rangle & \alpha \in \{\alpha_i\}. \end{cases}$$

Summing over all single-particle states,

$$\hat{N} = \sum_{\alpha} a_\alpha^\dagger a_\alpha, \quad \hat{N} |\alpha_1 \alpha_2 \dots \alpha_n\rangle = n |\alpha_1 \alpha_2 \dots \alpha_n\rangle.$$

- $\hat{N}$ is our first **one-body operator**: it acts on one single-particle state at a time.
- Most systems we study conserve particle number, and $[\hat{H}, \hat{N}] = 0$ is a useful consistency test of the formalism (and of computer programs!). In reactions such as (d, p) , states with different particle numbers are needed – second quantization handles them for free.

One-body operators: from coordinates to operators

Take $\hat{H}_0 = \sum_i \hat{h}_0(x_i)$, and expand the action of $\hat{h}_0$ in the single-particle basis,

$$\hat{h}_0(x_i)\psi_{\alpha_k}(x_i) = \sum_{\alpha'_k} \psi_{\alpha'_k}(x_i) \langle \alpha'_k | \hat{h}_0 | \alpha_k \rangle.$$

Acting with $\hat{H}_0$ on the product $\psi_{\alpha_1}(x_1) \cdots \psi_{\alpha_n}(x_n)$ replaces, in turn, each orbital α_k by α'_k weighted with $\langle \alpha'_k | \hat{h}_0 | \alpha_k \rangle$. Repeating for every permutation in the determinant and collecting the phases $(-1)^p$,

$$\hat{H}_0 |\alpha_1 \alpha_2 \dots \alpha_n\rangle = \sum_{k=1}^n \sum_{\alpha'_k} \langle \alpha'_k | \hat{h}_0 | \alpha_k \rangle |\alpha_1 \alpha_2 \dots \alpha'_k \dots \alpha_n\rangle.$$

The replacement $\alpha_k \rightarrow \alpha'_k$ inside a ket is precisely what $a_{\alpha'_k}^\dagger a_{\alpha_k}$ does:

$$|\alpha_1 \alpha_2 \dots \alpha'_k \dots \alpha_n\rangle = a_{\alpha'_k}^\dagger a_{\alpha_k} |\alpha_1 \alpha_2 \dots \alpha_k \dots \alpha_n\rangle.$$

One-body operators in second quantization

Inserting this and letting the sums run freely,

One-body operator, second quantized form

$$\hat{H}_0 = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle a_{\alpha}^{\dagger} a_{\beta}.$$

- $\hat{H}_0$ moves a particle from the single-particle state β to the state α , with amplitude $\langle \alpha | \hat{h}_0 | \beta \rangle$.
- Nothing here is specific to the kinetic energy: any particle-conserving one-body operator has this form – external fields, multipole operators, the number operator ($\langle \alpha | 1 | \beta \rangle = \delta_{\alpha\beta}$ gives back $\hat{N}$).
- All reference to coordinates, permutations and determinants is gone; the physics sits in the matrix elements $\langle \alpha | \hat{h}_0 | \beta \rangle$, computed once in the chosen basis.

A first consistency check, by pure algebra

Verify the expectation value between single-particle states $|\alpha_2\rangle = a_{\alpha_2}^\dagger |0\rangle$:

$$\langle\alpha_1|\hat{H}_0|\alpha_2\rangle = \sum_{\alpha\beta} \langle\alpha|\hat{h}_0|\beta\rangle \langle 0|a_{\alpha_1} a_{\alpha}^\dagger a_{\beta} a_{\alpha_2}^\dagger|0\rangle.$$

Anticommute annihilation operators to the right,

$$a_{\alpha_1} a_{\alpha}^\dagger a_{\beta} a_{\alpha_2}^\dagger = (\delta_{\alpha\alpha_1} - a_{\alpha}^\dagger a_{\alpha_1})(\delta_{\beta\alpha_2} - a_{\alpha_2}^\dagger a_{\beta}),$$

and every term still containing an operator dies on the vacuum:

$$\langle 0|a_{\alpha_1} a_{\alpha}^\dagger a_{\beta} a_{\alpha_2}^\dagger|0\rangle = \delta_{\alpha\alpha_1} \delta_{\beta\alpha_2} \implies \langle\alpha_1|\hat{H}_0|\alpha_2\rangle = \langle\alpha_1|\hat{h}_0|\alpha_2\rangle. \checkmark$$

Three lines. Remember this manoeuvre – *push annihilation operators to the right, collect the deltas* – it is tomorrow's whole subject, and Wick's theorem is its industrialisation.

Pause to think: the number operator and $\hat{H}_0$

Pause to think #3

(about 4 min – discuss with your neighbour)

1. Using only $\{a_\alpha^\dagger, a_\beta\} = \delta_{\alpha\beta}$ and $\{a_\alpha, a_\beta\} = 0$, show that $[\hat{N}, a_\gamma^\dagger] = +a_\gamma^\dagger$ and $[\hat{N}, a_\gamma] = -a_\gamma$.
2. Use this to argue, without any further computation, that $[\hat{N}, \hat{H}_0] = 0$ for $\hat{H}_0 = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle a_\alpha^\dagger a_\beta$.

Answer to pause #3

1. $[a_\alpha^\dagger a_\alpha, a_\gamma^\dagger] = a_\alpha^\dagger \{a_\alpha, a_\gamma^\dagger\} - \{a_\alpha^\dagger, a_\gamma^\dagger\} a_\alpha = \delta_{\alpha\gamma} a_\alpha^\dagger$, using $[AB, C] = A\{B, C\} - \{A, C\}B$. Summing over α gives $[\hat{N}, a_\gamma^\dagger] = a_\gamma^\dagger$; Hermitian conjugation flips the sign for a_γ .
2. Each term of $\hat{H}_0$ contains one creation and one annihilation operator, so it raises the count by one and lowers it by one: $[\hat{N}, a_\alpha^\dagger a_\beta] = (+1 - 1) a_\alpha^\dagger a_\beta = 0$. Every particle-conserving operator contains equally many $a^\dagger$'s and a 's – and this argument will apply verbatim to $\hat{H}_I$ tomorrow.

Summary of Thursday, and what comes on Friday

Today

- Week-34 reminder: variational principle, direct and exchange, energy functional – and the cost of first-quantized bookkeeping
- $a_{\alpha}^{\dagger}$, a_{α} ; antisymmetric states as operator strings on the vacuum
- $\{a_{\alpha}^{\dagger}, a_{\beta}^{\dagger}\} = \{a_{\alpha}, a_{\beta}\} = 0$, $\{a_{\alpha}^{\dagger}, a_{\beta}\} = \delta_{\alpha\beta}$
- $\hat{N} = \sum_{\alpha} a_{\alpha}^{\dagger} a_{\alpha}$ and $\hat{H}_0 = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle a_{\alpha}^{\dagger} a_{\beta}$

Friday

- The two-body interaction $\hat{H}_I$ in second quantization: results, with the derivations in section 3.2 of the book
- Normal ordering and contractions
- Two, three and four operators, done by hand
- Wick's theorem, its proof, and what it buys
- Exercise session (exercises 1 and 2 of the week-35 set)

Before Friday: run the first cells of `week35.ipynb` (the Fock-space matrices and the anticommutator checks), and redo pause #3 on paper – the identity $[AB, C] = A\{B, C\} - \{A, C\}B$ is used constantly.

Friday, lecture 3: The two-body interaction, normal ordering and contractions

Where we are

Thursday gave us the algebra,

$$\{a_{\alpha}^{\dagger}, a_{\beta}^{\dagger}\} = \{a_{\alpha}, a_{\beta}\} = 0, \quad \{a_{\alpha}^{\dagger}, a_{\beta}\} = \delta_{\alpha\beta},$$

and the one-body operator $\hat{H}_0 = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle a_{\alpha}^{\dagger} a_{\beta}$.

Today, second quantization only:

- the two-body interaction $\hat{H}_I$ in second quantized form – we **state the results** and leave the derivations to section 3.2 of the book;
- then the main course: **normal ordering**, **contractions**, and **Wick's theorem** with its proof (sections 3.5–3.11) – the machine that evaluates arbitrary operator strings.

The particle-hole formalism and the normal-ordered Hamiltonian (sections 3.3–3.4) come next week; Wick's theorem is stated here with respect to the particle vacuum $|0\rangle$.

The two-body interaction: the result

Starting from $\hat{H}_I = \sum_{i < j} V(x_i, x_j)$ and proceeding exactly as for $\hat{H}_0$ – expanding the action on products of orbitals, rewriting the replacement of a *pair* of orbitals as $a^\dagger a^\dagger a a$, and lifting the restriction on the pair sums – one finds

Two-body operator, second quantized form

$$\hat{H}_I = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma,$$

with all four sums running freely over the single-particle basis.

- Note the order of the annihilation operators: $a_\delta a_\gamma$, **reversed** relative to the ket labels of $\langle \alpha\beta | \hat{v} | \gamma\delta \rangle$.
- The derivation (three steps, each analogous to Thursday's one-body case) is in section 3.2 of the book; we do not repeat it here.

The two-body interaction: antisymmetrised form, and the full Hamiltonian

Using $\langle\alpha\beta|\hat{v}|\gamma\delta\rangle = \langle\beta\alpha|\hat{v}|\delta\gamma\rangle$ and the anticommutation rules, the same operator can be written with antisymmetrised matrix elements,

$$\hat{H}_I = \frac{1}{4} \sum_{\alpha\beta\gamma\delta} [\langle\alpha\beta|\hat{v}|\gamma\delta\rangle - \langle\alpha\beta|\hat{v}|\delta\gamma\rangle] a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma = \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{AS} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma.$$

The prefactor $-\frac{1}{2}$ or $\frac{1}{4}$ – tells which convention is in use. As a check (book, section 3.2), the pure operator algebra gives

$$\langle\alpha_1\alpha_2|\hat{H}_I|\beta_1\beta_2\rangle = \langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle - \langle\alpha_1\alpha_2|\hat{v}|\beta_2\beta_1\rangle = \langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle_{AS},$$

the direct-minus-exchange structure of week 34 – obtained without touching a determinant.

The Hamiltonian for the rest of the course

$$\hat{H} = \sum_{\alpha\beta} \langle\alpha|\hat{t} + \hat{u}_{\text{ext}}|\beta\rangle a_\alpha^\dagger a_\beta + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \langle\alpha\beta|\hat{v}|\gamma\delta\rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma, \quad (\text{antisymmetrised elements assumed}).$$

Pause to think: the factors and the operator order

Pause to think #4

(about 4 min – discuss with your neighbour)

1. Why must the annihilation operators appear in the reversed order $a_\delta a_\gamma$ rather than $a_\gamma a_\delta$? (Hint: check the sign of $\langle \gamma\delta | \hat{H}_I | \gamma\delta \rangle$ for the simplest two-particle case.)
2. Where does the change of prefactor $\frac{1}{2} \rightarrow \frac{1}{4}$ come from when we switch to antisymmetrised matrix elements?

Answer to pause #4

1. With the reversed order, $a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}$ applied to $|\gamma\delta\rangle = a_{\gamma}^{\dagger} a_{\delta}^{\dagger} |0\rangle$ removes γ first and then δ without picking up an extra transposition, so the diagonal matrix element of $\hat{v}$ enters with a $+$ sign. With $a_{\gamma} a_{\delta}$ every term would change sign, and the interaction energy of two particles would come out negative of what it should be. (Verified operator by operator in the notebook.)
2. The antisymmetrised element is itself a difference of two terms, and the operator string $a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}$ is antisymmetric under $\gamma \leftrightarrow \delta$ (and under $\alpha \leftrightarrow \beta$). Symmetrising the $\frac{1}{2}$ -form therefore counts every distinct contribution twice, and the prefactor must drop to $\frac{1}{4}$ to compensate.

The pattern – and its limit

Every explicit computation so far followed the same recipe:

1. push annihilation operators to the right with $a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p$;
2. let them die on the vacuum, $a_p |0\rangle = 0$;
3. collect the Kronecker deltas that fall out along the way.
 - For two operators: immediate.
 - For the four-operator overlap of two-particle states: an exercise in care.
 - For the **eight or more** operators of perturbation theory and coupled cluster theory: hopeless – the recursion branches at every step.

Wick's theorem

organises this entire calculation once and for all. We need two definitions first: **normal ordering** and **contractions**.

Normal ordering

A product of creation and annihilation operators is **normal-ordered** when all creation operators stand to the left of all annihilation operators. For an arbitrary product we define

$$N[xyz \cdots w] = (-1)^\kappa \text{ (the same operators, all creation operators to the left),}$$

where κ counts the interchanges of neighbouring operators needed to get there – each interchange of two fermion operators costs a minus sign. Two elementary cases:

$$N[a_p a_q^\dagger] = -a_q^\dagger a_p, \quad N[a_p^\dagger a_q] = a_p^\dagger a_q,$$

the first requiring one interchange, the second none.

The ordering of the creation operators among themselves (and of the annihilation operators among themselves) is a matter of convention, as long as the sign rule is applied consistently.

Why normal ordering is useful

The one property that matters

$$\langle 0|N[xyz\cdots w]|0\rangle = 0$$

for every non-empty product.

Whatever else it contains, a normal-ordered product has an annihilation operator on the far right or a creation operator on the far left, and either one kills the vacuum:

$$a_p|0\rangle = 0, \quad \langle 0|a_p^\dagger = 0.$$

A normal-ordered product therefore contributes *nothing* to a vacuum expectation value. The whole problem of evaluating $\langle 0|xyz\cdots w|0\rangle$ reduces to finding the pieces that are **not** normal-ordered.

Contractions

The **contraction** of two operators is the difference between their product and their normal-ordered product, indicated by a line joining the two,

$$\boxed{x y} \equiv xy - N[xy].$$

Since the normal-ordered piece has vanishing vacuum expectation value, a contraction is **just a number**,

$$\boxed{x y} = \langle 0|xy|0\rangle.$$

A number can be pulled outside any operator string. This is what will let us peel an operator product apart, pair by pair.

The four elementary contractions

There are four possible pairs of operators, and **only one survives**:

$$\overline{a_p a_q^\dagger} = \langle 0 | a_p a_q^\dagger | 0 \rangle = \delta_{pq},$$

$$\overline{a_p a_q} = \langle 0 | a_p a_q | 0 \rangle = 0,$$

$$\overline{a_q^\dagger a_p} = \langle 0 | a_q^\dagger a_p | 0 \rangle = 0,$$

$$\overline{a_p^\dagger a_q^\dagger} = \langle 0 | a_p^\dagger a_q^\dagger | 0 \rangle = 0.$$

Only an annihilation operator standing to the *left* of a creation operator contracts to something non-zero. This single fact – the statement that **the vacuum contains no particles** – carries all the bookkeeping that follows.

Signs: the crossing rule

When the two contracted operators are not adjacent, the contraction carries the sign of the interchanges needed to bring them together. For a set of contractions the total sign is

$$(-1)^\nu, \quad \nu = \text{number of crossings of the contraction lines},$$

since each crossing corresponds to one interchange of neighbouring fermion operators.

We shall use this rule constantly; it is verified numerically for arbitrary operator strings by `matching_sign` in `wick.py`, and in the notebook.

Pause to think: a three-operator warm-up

Pause to think #5

(about 4 min – discuss with your neighbour)

Expand the product $a_p a_q a_r^\dagger$ as a normal-ordered piece plus contraction terms, i.e. write down its Wick expansion by hand:

1. Which pairs can be contracted at all?
2. What signs do the surviving terms carry, and why?
3. What is $\langle 0 | a_p a_q a_r^\dagger | 0 \rangle$?

Answer to pause #5

1. Only annihilation-to-the-left-of-creation pairs survive: $(a_q, a_r^\dagger)$ and $(a_p, a_r^\dagger)$. The pair (a_p, a_q) contracts to zero.
2. Contracting adjacent $a_q a_r^\dagger$ costs no interchange: term $+\delta_{qr} a_p$. Contracting a_p with $a_r^\dagger$ requires moving them together past a_q – one interchange: term $-\delta_{pr} a_q$. Hence

$$a_p a_q a_r^\dagger = N[a_p a_q a_r^\dagger] + \delta_{qr} a_p - \delta_{pr} a_q.$$

3. Zero: an odd string leaves an uncontracted operator in every term, and $\langle 0|a_p|0\rangle = \langle 0|N[\dots]|0\rangle = 0$. (More on odd strings in a moment.)

Friday, lecture 4: Wick's theorem and its proof

Two operators

We now build the theorem up from the simplest cases, using nothing more than the anticommutation relations. For two operators the content of the theorem is the definition of the contraction, rearranged:

$$\overline{xy} = N[xy] + x y .$$

Take $x = a_p$, $y = a_q^\dagger$. The anticommutation relation gives

$$a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p = \delta_{pq} + N[a_p a_q^\dagger],$$

identifying the contraction as δ_{pq} . Taking the vacuum expectation value recovers the overlap of two single-particle states,

$$\langle p|q \rangle = \langle 0|a_p a_q^\dagger|0 \rangle = \delta_{pq}.$$

The three remaining combinations $a_p a_q$, $a_p^\dagger a_q$, $a_p^\dagger a_q^\dagger$ are already normal-ordered up to a sign, so their contractions vanish, and so do their vacuum expectation values.

Three operators: odd strings vanish

An odd number of operators can never be paired up completely: whatever contractions we carry out, at least one operator is left over, and

$$\langle 0|a_p|0\rangle = \langle 0|a_p^\dagger|0\rangle = 0.$$

Hence

$$\langle 0|xyz|0\rangle = 0$$

for every choice of the three operators. Directly, with $x = a_p$, $y = a_q^\dagger$, $z = a_r^\dagger$:

$$a_p a_q^\dagger a_r^\dagger = (\delta_{pq} - a_q^\dagger a_p) a_r^\dagger = \delta_{pq} a_r^\dagger - a_q^\dagger (\delta_{pr} - a_r^\dagger a_p),$$

and every surviving term contains a single unpaired creation operator acting on the vacuum.

Worth stating explicitly, because it is used constantly: **any vacuum expectation value of an odd number of operators is zero**. The same argument disposes of half the terms in the proof of the theorem.

Four operators, the hard way (1/2)

Four operators is the first genuinely interesting case: the overlap of two two-particle states,

$$|pq\rangle = a_p^\dagger a_q^\dagger |0\rangle, \quad \langle rs|pq\rangle = \langle 0|a_s a_r a_p^\dagger a_q^\dagger |0\rangle.$$

By anticommutation. Push a_r past $a_p^\dagger$ with $a_r a_p^\dagger = \delta_{rp} - a_p^\dagger a_r$:

$$a_s a_r a_p^\dagger a_q^\dagger = \delta_{rp} a_s a_q^\dagger - a_s a_p^\dagger a_r a_q^\dagger.$$

The first term is the two-operator case again,

$$\delta_{rp} a_s a_q^\dagger = \delta_{rp} (\delta_{sq} - a_q^\dagger a_s) \longrightarrow \delta_{rp} \delta_{sq},$$

the arrow denoting that we take the vacuum expectation value and drop the normal-ordered remainder.

Four operators, the hard way (2/2)

For the second term, anticommute twice more:

$$-a_s a_p^\dagger a_r a_q^\dagger = -a_s a_p^\dagger (\delta_{rq} - a_q^\dagger a_r) = -\delta_{rq} a_s a_p^\dagger + a_s a_p^\dagger a_q^\dagger a_r,$$

and the last term dies on the vacuum (annihilation operator on the far right). The survivor:

$$-\delta_{rq} a_s a_p^\dagger = -\delta_{rq} (\delta_{sp} - a_p^\dagger a_s) \longrightarrow -\delta_{rq} \delta_{sp}.$$

Collecting the two contributions

$$\langle rs|pq\rangle = \langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle = \delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq}.$$

Two-particle states are orthonormal up to precisely the antisymmetry expected of fermions – the overlap is the antisymmetrised product of single-particle overlaps. Four applications of the anticommutator, and constant vigilance about which terms survive.

Four operators, the easy way: contractions

The same answer can be read off without any algebra. Four operators can be paired in three ways.

1. Annihilation with annihilation, creation with creation – both contractions vanish:

$$\overbrace{a_s a_r} \overbrace{a_p^\dagger a_q^\dagger} = 0.$$

2. The nested pairing – no crossings, plus sign:

$$\overbrace{a_s a_r} \overbrace{a_p^\dagger a_q^\dagger} = +\delta_{sq}\delta_{rp}.$$

3. The crossed pairing – one crossing, minus sign (one line drawn below):

$$\overbrace{a_s a_r} \overbrace{a_p^\dagger a_q^\dagger} = -\delta_{sp}\delta_{rq}.$$

Adding the three: $\delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq}$ again. One pairing discarded on sight, two read straight off – **this economy is what the theorem buys.**

Wick's theorem

The pattern generalises. **Wick's theorem**: an arbitrary product of creation and annihilation operators equals its normal-ordered product plus the sum of all possible contractions,

$$\begin{aligned}xyz \cdots w &= N[xyz \cdots w] \\&+ \sum_{(1)} N[xyz \cdots w] \quad (\text{all single contractions}) \\&+ \sum_{(2)} N[xyz \cdots w] \quad (\text{all double contractions}) \\&+ \cdots + \sum_{[M/2]} N[xyz \cdots w],\end{aligned}$$

where $\sum_{(m)}$ runs over all ways of choosing m disjoint pairs to contract, each term carrying the crossing sign $(-1)^\nu$, and $[M/2]$ is the largest number of disjoint pairs that M operators admit.

Vacuum expectation values: fully contracted terms only

Take the vacuum expectation value of the theorem. Every term that still contains an uncontracted operator is normal-ordered and vanishes. Only the *fully* contracted terms survive:

Wick's theorem for vacuum expectation values

$$\langle 0 | xyz \cdots w | 0 \rangle = \sum_{[M/2]} N[xyz \cdots w] \quad (\text{fully contracted terms only}).$$

- For M even there are $(M - 1)!!$ such terms – one, three, fifteen, one hundred and five for $M = 2, 4, 6, 8$ – and most vanish on inspection because one of their contractions is of a vanishing type.
- For M odd there are none: odd strings give zero, as established.
- Each surviving term is a product of Kronecker deltas times $(-1)^{\text{crossings}}$.

Counting the work: the two routes compared

For strings of n annihilation followed by n creation operators ($M = 2n$), from the book (Fig. 3.1) and `wick.py`:

M	pairings $(M - 1)!!$	recursive calls	surviving terms $n!$
4	3	9	2
8	105	129	24
10	945	651	120
14	135 135	27 399	5 040

- The theorem does *not* win a race on raw counts – from $M = 10$ the recursion actually makes fewer calls, because branches that begin with $a^\dagger$ or end with a die immediately.
- What separates the routes: the pairings are known **in advance**, a purely combinatorial object; each can be inspected alone, most discarded at a glance, and the $n!$ survivors written down directly with their crossing signs. The recursion delivers one unsimplified answer at the very end and cannot be used as an ingredient in an algebraic argument.
- The ratio $(2n - 1)!!/n!$ – already 27 at $M = 14$ – is what makes hand calculation possible.

Why this matters beyond bookkeeping

Many-body connection

An n -body operator can connect only Slater determinants differing by at most n single-particle states – with a two-body force, at most two. A configuration-interaction (shell-model) Hamiltonian matrix is therefore **extremely sparse**: of the $\binom{n}{N}^2$ possible matrix elements, only a vanishing fraction are non-zero. That sparsity is what iterative eigensolvers such as the Lanczos algorithm exploit, and it is why the Hamiltonian matrix never has to be stored in full. The precise statement – the Slater-Condon rules – falls out of Wick's theorem in a few lines, once we apply the machinery.

Quantum computing connection

On a quantum computer the same strings $a^\dagger a$ and $a^\dagger a^\dagger a a$ become, through the Jordan-Wigner transformation (later in this course), strings of Pauli gates. Wick's theorem is the classical way of evaluating their expectation values; a quantum computer estimates them by repeated measurement instead.

Pause to think: six operators

Pause to think #6

(about 3 min – discuss with your neighbour)

Consider $\langle 0 | a_c a_b a_a a_d^\dagger a_e^\dagger a_f^\dagger | 0 \rangle$ ($M = 6$).

1. How many pairings does Wick's theorem enumerate?
2. How many of them survive the elementary contractions, and why exactly that many?
3. What does each surviving term look like?

Answer to pause #6

1. $(6 - 1)!! = 5 \cdot 3 \cdot 1 = 15$.
2. A non-vanishing contraction must join one of the three annihilation operators (all standing left) to one of the three creation operators (all standing right). Every such complete assignment is a permutation of three objects: $3! = 6$ survivors.
3. A product of three deltas with the parity of the permutation as sign, e.g. $+\delta_{ad}\delta_{be}\delta_{cf}$ for the nested pairing and $-\delta_{ae}\delta_{bd}\delta_{cf}$ after one transposition – exactly the six terms of a 3×3 determinant $\det(\delta_{i\mu})$. Wick's theorem knows about determinants without ever being told.

Proof of Wick's theorem: the strategy

The proof is by induction on the number of operators M .

- The case $M = 2$ is the definition of the contraction rearranged – already established.
- The induction step requires one preparatory result, which does all the real work:

Lemma: appending one operator on the right

Let $x, y, z, \dots, w$ be operators and Ω one further creation or annihilation operator. Then

$$N[xyz \cdots w] \Omega = N[xyz \cdots w \Omega] + \sum_{(1)\Omega} N[xyz \cdots w \Omega],$$

where $\sum_{(1)\Omega}$ runs over all single contractions *involving* Ω – the contractions of Ω with each of $x, y, z, \dots, w$ in turn.

In words: multiplying a normal-ordered product by one more operator costs exactly one round of contractions with that operator.

Proof of the lemma: the three easy cases

- (i) Ω is an annihilation operator. Appending it on the far right cannot disturb the normal ordering, so the left-hand side already equals $N[xyz \cdots w \Omega]$. At the same time every contraction with Ω vanishes: a non-zero contraction needs an annihilation operator to the *left* of a creation operator, and Ω stands to the right of everything. Both extra terms are absent – the lemma holds trivially.
- (ii) All of $x, y, \dots, w$ are annihilation operators, Ω too. A special case of (i).
- (iii) Ω is a creation operator. Then it suffices to prove the lemma when all of $x, y, \dots, w$ are annihilation operators: any creation operator inside the product already stands to the left of Ω 's eventual position, and its contraction with Ω vanishes,

$$\overline{a_i^\dagger a_j^\dagger} = 0,$$

so creation operators in the string contribute neither to the reordering nor to the contraction sum, and are carried along untouched.

Proof of the lemma: the case that works

- (iv) It remains to move a creation operator Ω leftwards through annihilation operators $x, y, \dots, w$, one at a time. Each step uses

$$a_j \Omega = \delta_{j\Omega} - \Omega a_j,$$

producing two pieces:

- a **Kronecker delta** – precisely the contraction of a_j with Ω , carrying the sign of the operators Ω has crossed so far;
- a term in which Ω has moved **one place to the left**.

Iterating until Ω has passed all of them: the accumulated delta terms are exactly $\sum_{(1)\Omega} N[xyz \cdots w \Omega]$, and the final term – Ω to the left of every annihilation operator – is exactly $N[xyz \cdots w \Omega]$. $\square$

The lemma is nothing but the push-to-the-right recipe, run once and recorded as an identity.

The lemma at work

For $M = 1$:

$$N[a_1] a_\ell^\dagger = N[a_1 a_\ell^\dagger] + N \left[\overbrace{a_1 a_\ell^\dagger} \right] = -a_\ell^\dagger a_1 + \delta_{1\ell},$$

which is $a_1 a_\ell^\dagger = \delta_{1\ell} - a_\ell^\dagger a_1$ once more. For $M = 2$:

$$a_0 N[a_1] a_\ell^\dagger = N[a_0 a_1 a_\ell^\dagger] + N \left[\overbrace{a_0 a_1 a_\ell^\dagger} \right] + N \left[\overbrace{a_0 a_1 a_\ell^\dagger} \right],$$

that is $N[a_0 a_1 a_\ell^\dagger] + \delta_{1\ell} N[a_0] - \delta_{0\ell} N[a_1]$, the last sign from carrying the contraction past a_1 – the three-operator expansion of pause #5, now derived rather than guessed.

The induction step

Assume the theorem holds for M operators. Multiply both sides from the right by Ω and apply the lemma to every term:

- the leading term gives $N[xyz \cdots w] \Omega = N[xyz \cdots w \Omega] + \sum_{(1)\Omega} N[xyz \cdots w \Omega]$;
- each m -fold contracted term gives

$$\sum_{(m)} N[xyz \cdots w] \Omega = \sum_{(m) \neq \Omega} N[xyz \cdots w \Omega] + \sum_{(m)\Omega} N[xyz \cdots w \Omega],$$

splitting the m -fold contractions of $M + 1$ operators into those not involving Ω and those in which one contraction ends on Ω .

But an m -fold contraction of the $M + 1$ operators *either involves Ω or it does not*, so the two sums recombine into $\sum_{(m)}$ over all $M + 1$ operators. Collecting terms by the number of contractions reproduces the theorem with $M \rightarrow M + 1$. $\square$

A remark on parity, and where we stand

If $M + 1$ is odd, the deepest sum still leaves one operator uncontracted; on taking the vacuum expectation value those terms vanish, and we recover: *vacuum expectation values of odd strings are zero*. For $M + 1$ even the deepest sum consists of the fully contracted terms, and the boxed vacuum-expectation-value formula follows.

What we now own

An algorithm – enumerate pairings, discard vanishing contractions, attach crossing signs – that evaluates *any* vacuum expectation value of creation and annihilation operators, with a proof that it equals the anticommutation route, and a program (`wick.py`) that checks both.

Many-body connection

Next week we upgrade the vacuum: normal ordering and contractions *relative to a reference determinant* (the Fermi sea), the particle-hole formalism, and the generalised Wick theorem for products of normal-ordered groups – the form in which the theorem is used in Hartree-Fock, perturbation theory and coupled cluster calculations.

Pause to think: where was Ω 's nature used?

Pause to think #7

(about 3 min – discuss with your neighbour)

1. In the induction step we multiplied by Ω *on the right*. Why is it enough to prove the theorem this way – what about products that grow to the left?
2. Which single ingredient of the whole proof would fail for *bosons*, and what would replace it?

Answer to pause #7

1. Every product of $M + 1$ operators *is* a product of its first M operators times its last one, so right-multiplication reaches all strings; induction starting from $M = 2$ covers them all. (One can equally run the induction leftwards – the lemma has a mirror image with $\langle 0|\Omega$ -type arguments.)
2. The sign bookkeeping: for bosons the elementary relation is a *commutator*, $[b_p, b_q^\dagger] = \delta_{pq}$, interchanges cost no sign, and the crossing rule becomes trivial (+1 always). Wick's theorem holds for bosons too, with contractions $\langle 0|b_p b_q^\dagger|0\rangle = \delta_{pq}$ and no signs – the fermionic $(-1)^\nu$ is the only difference.

Exercise session: week 35

The two exercises of the week-35 set (chapter 2 of the lecture notes, where full answers are worked out) drill the Slater-determinant machinery of the reminder lecture, and build our first, very simple model of full configuration interaction theory.

Exercise 1: expectation values with a Slater determinant

Consider the fermion Slater determinant ansatz

$$\Phi_{\lambda}^{\text{AS}}(x_1 x_2 \dots x_N; \alpha_1 \alpha_2 \dots \alpha_N) = \frac{1}{\sqrt{N!}} \sum_p (-1)^p \hat{P} \prod_{i=1}^N \psi_{\alpha_i}(x_i),$$

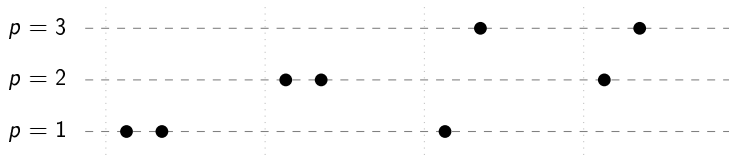
with the sum over all $N!$ permutations and $\{\psi_{\alpha_i}\}$ an orthonormal single-particle basis.

- Write out Φ^{AS} for $N = 3$.
- Show that $\int dx_1 \dots dx_N |\Phi_{\lambda}^{\text{AS}}(x_1 \dots x_N; \alpha_1 \dots \alpha_N)|^2 = 1$.
- Define a general one-body operator $\hat{F} = \sum_i^N \hat{f}(x_i)$ and a general two-body operator $\hat{G} = \sum_{i>j}^N \hat{g}(x_i, x_j)$, with $\hat{g}$ invariant under interchange of the coordinates of particles i and j . Calculate the matrix elements $\langle \Phi_{\alpha_1 \alpha_2}^{\text{AS}} | \hat{F} | \Phi_{\alpha_1 \alpha_2}^{\text{AS}} \rangle$ and $\langle \Phi_{\alpha_1 \alpha_2}^{\text{AS}} | \hat{G} | \Phi_{\alpha_1 \alpha_2}^{\text{AS}} \rangle$ for a two-particle Slater determinant. Which properties do you expect these operators to have, in addition to permutation symmetry?
- Repeat the calculation for a general N -particle Slater determinant.

Exercise session: week 35 (continued)

Exercise 2: a three-level model

A simple three-level problem – our first model of a many-fermion system, and of what we shall later call full configuration interaction theory. The particles are fermions; each single-particle state $p = 1, 2, 3$ is doubly degenerate (think spin up and down), the level spacing is constant with value d , and the first state has energy d .



Four possible two-particle states: filled circles are occupied single-particle states. The two leftmost configurations, with both particles in the same level, are the ones retained in parts b) and c).

Exercise session: week 35 (continued)

Exercise 2, the tasks

- a) How many two-particle Slater determinants can we construct in this space?
- b) Limit the system to the two lowest orbits, $p = 1, 2$, and two particles. Write the Hamiltonian as $\hat{H} = \hat{H}_0 + \hat{H}_I$ with $\hat{h}_0 \psi_{p\sigma} = p d \psi_{p\sigma}$, σ the spin quantum number. Assume that the only two-particle states that can exist are those in which both particles occupy the same level p , and that all two-particle matrix elements of $\hat{H}_I$ have the constant value $-g$. Show that the Hamiltonian matrix can be written as

$$\begin{pmatrix} 2d - g & -g \\ -g & 4d - g \end{pmatrix},$$

and find its eigenvalues and eigenvectors. What is the admixture of the state with two particles in $p = 2$ to the wave function with two particles in $p = 1$? Discuss the result in terms of a linear combination of Slater determinants.

- c) Add the possibility that the two particles occupy $p = 3$, and find the Hamiltonian matrix, its eigenvalues and eigenvectors, still allowing only two-particle states with both particles in the same level. The 3×3 matrix may be diagonalised numerically – see the last section of `week35.ipynb`.

Why this little model matters

The three-level model catches several birds with one stone:

- it shows how **linear combinations of Slater determinants** are built, and how the coefficients are interpreted as admixtures to a given state;
- the two-particle states above $p = 1$ are **excitations** of the ground-state configuration, and the reliability of the single-determinant ansatz depends on the ratio of the interaction strength g to the level spacing d ;
- it is a schematic ansatz for **pairing correlations** – and hence for superfluidity and superconductivity in fermionic systems.

Many-body connection

We will meet this model again: it is the ancestor of the pairing Hamiltonian used later in the course, and rewriting it in the second-quantization language of this week – pair operators $P_p^+ = a_{p\uparrow}^\dagger a_{p\downarrow}^\dagger$ acting on the vacuum – is a perfect first application of the operator algebra. Try writing $\hat{H}_0$ and $\hat{H}_I$ of exercise 2b) in terms of $a_{p\sigma}^\dagger$ and $a_{p\sigma}$.

What to do with the notebook

WeeklySlides/week35.ipynb contains, in this order:

1. Fermion creation and annihilation operators as matrices in a finite Fock space; all three anticommutation relations verified for every pair of states.
2. Slater determinants as operator strings: antisymmetry signs and the Pauli principle from the matrices.
3. The number operator and $\hat{H}_0 = \sum \langle \alpha | \hat{h}_0 | \beta \rangle a_\alpha^\dagger a_\beta$: eigenvalues, $[\hat{N}, \hat{H}_0] = 0$, and the week-34 sum over occupied levels recovered.
4. The two-body operator in both the $\frac{1}{2}$ - and the $\frac{1}{4}$ -form; the operator-order and prefactor checks of pause #4; $\langle ij | \hat{H}_I | ij \rangle = \langle ij | \hat{v} | ij \rangle_{\text{AS}}$.
5. Normal ordering and the four elementary contractions, checked as matrix identities.
6. Wick's theorem by brute-force anticommutation and by enumeration of pairings (a compact version of wick.py), verified to agree for $M = 2, \dots, 8$.
7. The cost table of the slides recomputed: pairings, recursive calls, survivors.
8. The overlap $\langle rs | pq \rangle$ and the six-operator determinant structure of pause #6.
9. The three-level model of exercise 2: the 2×2 and 3×3 Hamiltonian matrices diagonalised numerically, and the ground-state admixtures as functions of g/d .

Play with it: change the number of orbitals, the matrix elements, the operator strings.

Summary of week 35

The formalism

- Antisymmetric states are strings of creation operators on the vacuum.
- All antisymmetry lives in $\{a_\alpha^\dagger, a_\beta^\dagger\} = \{a_\alpha, a_\beta\} = 0$, $\{a_\alpha^\dagger, a_\beta\} = \delta_{\alpha\beta}$.
- $\hat{H}_0 = \sum \langle \alpha | \hat{h}_0 | \beta \rangle a_\alpha^\dagger a_\beta$;
 $\hat{H}_I = \frac{1}{4} \sum \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{AS} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma$.

The theorem

- Normal-ordered products have vanishing vacuum expectation values.
- A contraction is a number; only $a_p a_q^\dagger \rightarrow \delta_{pq}$ survives.
- Wick: product = normal-ordered + all contractions; in $\langle 0 | \cdots | 0 \rangle$ only fully contracted terms remain, with signs $(-1)^{\text{crossings}}$.
- Proof: one lemma (append an operator) plus induction.

Next week, and a closing remark

Next week (week 36)

The particle-hole formalism: a reference determinant as the new vacuum, quasiparticle operators, normal ordering relative to the Fermi sea, the normal-ordered Hamiltonian, and Wick's generalised theorem. Reading: chapter 3 of the lecture notes, sections 3.3–3.4 and 3.12–3.14.

This week's theorem was stated for the empty vacuum $|0\rangle$, where it reproduces – effortlessly – everything week 34 obtained from determinants. Its real power appears when the vacuum is replaced by a filled Fermi sea: then contractions of *hole* states survive too, expectation values in the reference determinant become one-line computations, and the direct and exchange terms of Hartree-Fock, the diagrams of perturbation theory and the amplitude equations of coupled cluster all fall out of the same machine. That is where we go next.